BALAKUMARAN K

TVS Credit Services, Chennai

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PROFESSIONAL SUMMARY

Data Science professional with 5 years of experience in solving complex business problems using data-driven approaches. Expertise in customer segmentation, behavioral analytics, and personalized product recommendations. Proven experience working across banking / finance and engineering domains, building and deploying models to drive strategic decisions. Committed to optimizing workflows and delivering measurable business impact.

EXPERIENCE

Data Analyst — **Chennai**
TVS Credit Services — **June 2024 – Present**

- Spearheaded Data Enrichment initiatives by partnering with external data providers to source alternative datasets and build POCs, enhancing customer behaviour tracking and market prediction capabilities.
- Drove reduction in off-us leakage by building a customer insights dashboard and leading data-driven performance tracking initiatives.
- Accelerated data processing by architecting an end-to-end ETL pipeline for bureau data, while managing the organization's prospect base.
- Delivered measurable business impact by building propensity models generating ₹25 crore in FY26 book value and deploying a real-time risk trigger system that improved collections by 9%.

Software Engineer – Data Science — **Chennai**
Stellantis / Hindujatech Pvt. Ltd. — **Oct 2022 – June 2024**

- Applied time series analysis, regression, and statistical techniques on vehicle testing data to develop ML models, reducing reliance on physical testing in automobile validation workflows.
- Conducted R&D on unexplored datasets without existing simulation frameworks, developing new prediction methodologies for vehicle performance analysis.
- Accelerated team capability by rapidly upskilling in MATLAB and building expertise in model development and impact analysis.
- Mentored and trained a Graduate Engineer Trainee at Stellantis on Python, statistics, and machine learning, while leading weekly technical discussions and tracking team learning progress.
- Strengthened team collaboration and knowledge sharing by coordinating technical discussions and monitoring skill development across team members.

ML Data Associate — **Chennai**
Amazon India Development Centre — **Jan 2019 – Sep 2020**

- Processed and transformed raw data into structured inputs for NLP systems through transcription, annotation, and prioritization of critical and non-critical data, improving data quality.
- Promoted from transcription to verification role within 3 months based on strong performance and high scores in internal assessments.
- Mentored new joiners and led three quality improvement initiatives to improve process efficiency and data accuracy.

PROJECTS

- **Propensity Model for Used Car and Personal Loan:**
  Business Objective – Unnecessary as well as mismatched product campaign leads to bad publicity, wasted effort, time and cost associated with each call. A propensity to buy model would thereby help in streamlining and driving a targeted campaign.
  Approach – Lead the team creating a propensity model using historical bureau data, the past DPD performance and frequency of enquiry.
  Delivery & Results – A final lightgbm model was developed with top 10 deciles for propensity to convert. The Used Car Conversion from prospects spiked from 20 a month to 80 on an average every month.
- **IRR Regression Model for Interest Rate Prediction:**
  Business Objective – Offering higher or incompatible interest rates to customer leads to dissatisfaction, poor onboarding experience and wasted campaign efforts.
  Approach – A three layered approach was taken looking at RBI repo rate, past performance / interest rate purchase of the prospect and the pin code level aggregated interest rates for the prospects / customers in the region.
  Delivery & Results – Heuristic + Regression Model was used to predict the ideal interest rate. The best of the heuristic and the ml model was chosen to drive campaign. The project has been deployed and the business impact is under study.
- **Automated Compliance tool using LLM:**
  Business Objective – RBI's DAKSH portal issues numerous, text-heavy circulars and notifications, making it difficult for teams to effectively interpret and implement compliance requirements. The objective was to provide concise summaries to respective departments for better compliance.
  Delivery & Results – Open-source MISTRAL LLM model was used to summarize the textual data. Inherently understand the intended department and send an automated mail to the concerned stakeholders in the department.

SKILLS

Programming	**:** Python, SQL
Data Processing & Analytics	**:** Pandas, Numpy, Statistics, Microsoft Excel, Tableau
GenAI Tools	**:** Ollama, LangGraph, OpenAI API, FAISS
Cloud	**:** Amazon Web Services, Oracle Cloud Infrastructure
Soft Skills	**:** Team lead, Effective at taking ownership and delegating tasks, Effective communication and conversation driver.

EDUCATION

Post Graduate Program in Data Science and Engineering, — **Jan 2022 – Aug 2022**
Great Lakes Institute of Management — *Chennai*

B. E. Mechanical Engineering, — **Aug 2012 – May 2016**
CSI College of Engineering — *Ketti, The Nilgiris*

CERTIFICATIONS

AWS Certified AI Practitioner: https://www.credly.com/badges/4b5333bc-d741-4eb1-9ad9-fe75ef91b99a